# ME 5309 – Mid-Semester Examination

Answer Booklet

**Instructions:** Attempt any three questions. Each question carries 10 marks. Show working where appropriate.

## Question 1 [10 marks]

### Q1(a) [4 marks]

$$f_x = 3200 pixels; fieldofview = 160mm; 1pixel = 0.125mm$$

### Q1(b) [3 marks]

ADC step = 5/4096 = 1.2207 mV. I divide by 0.1 mV/N, giving ΔF = 12.207 N.

### Q1(c) [3 marks]

$$F = e^(AT) = e^(-0.2) = 0.818731$$

## Question 2 [10 marks]

### Q2(a) [2 marks]

At equilibrium the velocity should be zero. The torque and gravity torque must balance.

### Q2(b) [2 marks]

The Jacobian has $A_{12}$ = 1 and $A_{22}$ = -b/J = -2.5. The input matrix is B = [0, 50]$^T$.

### Q2(c) [4 marks]

At rest, $\dot{\theta} = \ddot{\theta} = 0$, so the torque balance is Kp(θd-θss) = mgℓ sin θss. With Kp = 50, θd = π/2 and mgℓ = 2.4525, solve 50(π/2-θss) = 2.4525 sin θss. This gives θss approximately 1.52181 rad = 87.193°, so the error is 90°-87.193° = 2.807°; it is not zero.

Rearranging the steady-state equation gives $50(\pi/2 - \theta_{ss}) = 2.4525 \sin\theta_{ss}$. I used radians in the equation. Solving numerically gives $\theta_{ss} \approx 1.522$ rad $\approx 87.19°$, so the gap from $90°$ is about $2.81°$.

### Q2(d) [2 marks]

## Question 3 [10 marks]

### Q3(a) [2 marks]

Finite differencing amplifies encoder noise, making the velocity estimate noisy.

### Q3(b) [2 marks]

Tustin uses s = (2/T)(z-1)/(z+1).

### Q3(c) [4 marks]

### Q3(d) [2 marks]

The observer is better and the filter is safer in some cases.